

Facial Emotion Detection Using Deep Learning: A Survey

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Abstract—The long history of facial expression analysis has influenced current research and public interest. The scientific study and comprehension of emotion are credited to Charles Darwin's 19th-century publication *The Representation of the Sentiment in Man and Animals* (originally published in 1872). As Recognition of human emotions from images is one of the utmost important and difficult societal connection study assignments. One advantage of using a deep learning strategy is its independence from human intervention while undertaking feature engineering. This approach involves an algorithm that scans the data for features that connect, then combines them to promote quicker learning without being explicitly told to. Deep learning (DL) based emotion detection outperforms traditional image processing methods in terms of performance. In this analytical study, the creation of an artificial intelligence (AI) system that can recognize emotions from facial expressions is presented. It discusses the various techniques for doing so, which generally involve three steps: face uncovering, feature extraction, and sentiment categorization. This study describes the various existing solutions and methodologies used by the researchers to build facial landmark interpretation. The Significance of this survey paper is to analyze the recent works on facial expression detection and distribute better insights to novice researchers for the upgradation in this domain.

Keywords—Facial Emotion, Deep Learning, Convolutional Neural Network

I. INTRODUCTION

Due to the proliferation of surveillance cameras in recent years, gesture expression detection has become a crucial area of research. It assists us in expressing our internal emotional states and recognizing other people's emotions and behavior. In response to the increasing requirement for robotic systems to assist humans in a centralized manner, researchers are deciding whether to integrate robots in various workspaces. These robots must, however, understand human emotions to interact naturally. To achieve perfect FER system performance, numerous FER researchers are diligently working to improve new FER algorithms, but few have published their findings. Extended Cohn and Kanade (CK+) datasets and FER 2013 are among the ones that most academics advise using to test their specialized FER algorithms. Well-thought and quite often used sets of data include FER 2013 and CK+. Sadly, whether dataset offers

much statistics, especially in contrast to the 60,000 object images in the CIFAR-10 dataset, employed for testing and training.

A significant proportion of algorithms have 99% predictive performance nowadays, thanks to recent advancements. Relevant facial photos must be purged from the FER 2013 collection to lessen pointless computation. Insufficient training and assessment sets of data and an uneven distribution in every catalog of facial pictures resulted from removing any facial images from the dataset, which had many available facial photos. We integrate with various existing datasets or gather more data sources should include extra frontal pictures for testing and training to substitute the facial images that have been redacted.

The majority of facial expression cues, according to observations, are provided by the region around the mouth, eyes, and cheeks, with the other parts of the face supporting this region and highlighting the emotional movements. One of the crucial jobs of any image-based recognition system is picture analysis. A face picture serves as the system's input in a FER system. If F is an image of $M \times N$ dimensions, it could be colored or in grayscale. A cutting-edge technique of Utilizing face expression recognition, deep learning uses neural networks to analyze and decipher human emotions from facial expressions.

In this survey research article, the importance of CNN is that it is a modern architecture used in the deep learning domain and is especially known for providing much-optimized results in the field of gesture detection.

With this method, a model is trained using a sizable dataset of pictures of people's faces that have been professionally labeled with their emotions. The model then develops the ability to recognize patterns in the images and infer the emotions being expressed. This technique is quite accurate and has innumerable innovations, particularly in the areas of study of psychology, marketing, and security. Additionally, those with disabilities or communication difficulties can use this technology. The strategy by using the range of data that had been mentioned with a main objective, whereby the exactness has been procured by the procedure accelerates forward into nominating the information, analyzing the data, training the data, selecting the suitable template that would

yield high accuracy and keep on going to be adequate to the challenge and retrieving the highlights from the provided image to generate a list to facilitate the prototype. Afterward when the framework will be using initiation and generalization to react appropriately. In this paper, the main objective is to address all the existing solutions used for human face sentiment identification and give a suitable awareness to future researchers for advancing face expression detection. In the remainder of the paper section 2 is described, which states the most previous recognition of facial emotion

research that has been done. Section 3 discusses the conclusion of the paper in which the best method and the future scope of work are being determined.

II. RELATED WORK

To better understand all the related works, table 1 describes various parameters used by researchers in their work. Table 1 is defining the parameters such as Author names, identification rate, sets of data used, and their contribution.

TABLE I. COMPARISON OF DIFFERENT METHODOLOGIES AND THEIR ACCURACY

Ref. and year	Author's Name	Method	Recognition accuracy	Datasets Used	Number of Expressions	Advantages and Major Contribution
[1]2020	Akriti Jaiswal, A. Krishnama Raju, Suman Deb.	CNN, Keras	70.14% and 98.65%	FER and JAFFE	7	On the different datasets, it is garnering cutting-edge impacts and is superior to earlier models concerning the consequences of emotion classification.
[2]2020	Wafa Mellauk, Wahida Handouzi	CNN, LSTM	Very high (Exact number not specified)	MultiPie, MMI, GEMEP, FERA, SFEW	7	Examined the most recent FER developments. With the help of this, the ability of machines to understand emotions will increase.
[3]2016	Daniel Llatas, Spiers	CNN, Tensorflow	80%	Extended Cohn-Kanade and Affectiva-MIT Facial Expression Dataset	6	Deep learning techniques have been studied to categorize facial emotions from static facial images in this paper.
[4]2016	Ali Mollahosseini, David Chan Mohammad H. Mahoor	CNN, GoogLeNet and AlexNet Architecture	90.5%, 62.1% and 81.7%	FERA, SFEW and FER2013	7	While requiring fewer operations to train the network, it is improving classification accuracy in cross-database measurement circumstances and pertaining individuality.
[5]2020	Pranav E., Suraj Kamal, Satheesh Chandran C., Supriya M.H.	Jupyter Notebook, Keras, TensorFlow, Scikit Learn, Matplotlib, Seaborn, and Adam Optimizer	78.04%	Not Specified	5	Indicating that the model is having the most precise conform and is pertinent to the records, it has comparable training and validation accuracy.
[6]2015	Yunnan Li and Edmund Y. Lam	Viola-Jones algorithm, Gabor Wavelets, KPCA	96.80%	Extended Cohn-Kanade Dataset	6	When combined with Kernel PCA, fitting a normal distribution to the image histogram results in a higher recognition rate than a traditional histogram. equalization and linear PCA.
[7]2014	Yadan Lv, Zhiyong Feng, Chao Xu	DBN	90.47% and 91.11%	JAFFE and Extended Cohn-Kanade	7	To avoid changing the weighted function on various faces, the feature of parsed components is treated equally. Deep architectures that have been pre-trained with stacked autoencoders assess the efficiency of these components.
[8]2017	Abir Fathallah, Lotfi Abdi, Ali Douik	CNN, DNN, VGG	99.33%, 93.33%, and 87.65%	Ck+, RaFD, and MUG	5	Four convolutional layers help compensate for this; the very last layer is then accompanied by a completely connected layer first three are followed by max pooling. Results and recognition rates demonstrate that this method performs better than cutting-edge techniques.
[9]2015	Xiaoming Zhao, Xugan Shi, Shiqing Zhang	RBN, DBN	90.95% and 98.75%	JAFFE and Extended Cohn-Kanade	7	This study proposes a unique approach to recognizing facial gestures. that combines DBNs and MLP (abbreviated as DBNs + MLP).
[10]2022	Saiyed Umer, Ranjeet Kumar Rout, Chiara Pero, Michele Nappi	CNN, Tree-Structured Part Model (TSPM)	82.79%, 94.33% and 97.69%	KDEF, GENKI-4K, CK+	7	By fine-tuning the hyper-parameters, the data augmentation techniques not only address the overfitting issue but also improve the proposed system's performance.
[11]2021	Yahia Said, Mohammad Barr	FS-CNN, TensorFlow, CUDA	94.90%	Celebi, UMD	7	The proposed system performs better thanks to the data augmentation techniques, which also address the issue of overfitting by optimizing the hyper-parameters.
[12]2021	Xiaohui Tan and YongAn Huang	sEMG-Based FER Algorithm	85%	sEMG	5	The suggested system the obfuscation of appropriate AU tagging calls for knowledge and time and instead relies on underlying data AU to quantify AU.
[13]2021	Jung Hwan Kim, Alwin Poulouse, and Dong Seog Han	CNN, PyFER, MobileNet, Inception V.1, ResNet 50, and the Xception algorithm	77.94%	FER 2013, CK+ and SPL	Not Specified	To Discard any irrelevant information, rearrange This paper recommended a FIT machine for the FER dataset manager to consolidate all different datasets, acquire a new dataset, and incorporate all different data sources.

[14]2021	M. Kalpana Chowdary, Tu N. Nguyen, and D. Jude Hemanth	VGG 19, Resnet 50, MobileNet and Inception V3	96%, 97.7%, 98.5% and 94.2%	CK +	7	These networks will eventually be used to detect emotions in EEG signals and speech.
[15]2021	Shrey Modi, Mohammed Husain Bohara	CNN, Tensorflow	High (Comparatively)	Not Specified	3	Processing enables people with autism spectrum disorder (ASD) to access PCs.
[16]2021	Harisu Abdullahi Shehu, Md. Haidar Sharif and Sahin Uyaver	ResNet model, Haar Cascade (HC) classifier	81.68%, 98.78% and 96.67%	RAF, CK+, and KDEF	Not Specified	The HC was able to achieve quick convergence on the CK+ and KDEF databases. Although label smoothing can improve clustering results, it did not increase the accuracy of some databases.
[17]2021	Afshan Jamil, Serestina Viriri	CNN, ELM, Viola Jones and Algorithm	86.5%, 96.8% and 62.5%	Cohn- Kanade, JAFFE and FER2013	Not Specified	Without iteratively updating the weights, it decided to take a neural model and decoction the high-level features for elevated portrayal advantage., which reduces the computational time complexity.
[18]2021	Shervin Minaee, Mehdi Minaei, and Amirali Abdolrashidi	CNN	70.02%, 92.8%, 98.0% and 99.3%	FER2013, JAFFE, CK+ and FERG	7	This study proposes a novel attentional convolutional network-based framework for facial expression recognition.
[19]2020	Aya Hassouneh, A.M. Mutawalla and M. Murugappan	CNN, Lucas-Kanade optical flow algorithm	99.81%	Not Specified	6	The system aims to assist people who are along with supporting children with autism, they are also physically disabled (deaf, dumb, and bedridden). understand other people's emotions.
[20]2020	Amil Khanzada, Charles Bai and Ferhat Turker Celepcikav	Fine Tuning ResNet50, Fine Tuning SeNet50, Fine Tuning VGG16	75.80%	FER2013, JAFFE, Mobile WebApp Dataset, CK+	7	This paper overcame data mismatch issues by creating a training dataset and fine-tuning its architecture to run on-device with a minimum amount of memory, disc space, and computational power.
[21]2020	Md. Forhad Ali, Mehenag Khatun and Nakib Aman Turzo	Viola-Jones cascade object detectors, HOG, SVM	97%	CSV file	7	This has introduced decision-tree techniques that assist in determining which emotions have a high and low percentage. The majority of emotions are as accurate as they can be. In addition, emotions have a low probability of existing in low percentages.
[22]2021	M. A. H. Akhand, Md. Abdus Samad Kamal	CNN, DCNN, FER	98.78% and 99.52%	KDEF and JAFFE	7	Modern research is interoperable with more exhaustive commercial companies, particularly the effectiveness with profile stances. applications, like patient monitoring in hospitals or security surveillance.
[23]2021	H. Sikkandar and R. Thiagarajan	DCNN, Cat Swarm Optimization	96.76%, 94.59%, 95.34% and 96.28%	JAFFE, CK+, Pie dataset, and Real-world images	Different no. of expressions for different datasets	To increase the proposed system's overall accuracy and retrieval speed, this framework incorporates pre-processing, components identified, semantic segmentation (DCNN), variable selection (ICSO), and scene understanding (SVM, NN).
[24]2021	Karnati Mohan, Ayan Seal, Ondrej Krejcar, and Anis Yazid	CNN, FER-net, RELU	79%, 97%, 98%, 89% and 82%	FER2013, JAFFE, CK+, KDEF and RAF	7	In comparison to modern models, the suggested model is straightforward and superior in terms of accuracy and execution time.
[25]2020	Kuan Li, Yi Jin, Muhammad Waqar Akram, Ruize Han, and Jiongwei Chen	CNN, Momentum optimizer, Xavier initializer, cross	97.18% and 97.65%	CK+ and JAFFE	7	With the proposed rotation method, the amount of data is significantly increased while the suggested face-cropping method keeps useful facial information and eliminates useless regions.

After discussing recent works happened for facial expression identification the most important inferential point has been drawn that many researchers used convolutional and transfer learning-based deep learning algorithms in different kinds of available datasets. However, the transfer learning models receive the most optimized results. Furthermore, there are several advantages to transferring learning, but the largest ones include shorter training times, generally greater neural network performance, and less data requirement.

III. CONCLUSION

In this survey article, we examined the issue of facial expression detection, which has recently gained significant relevance in the field of HCI. We also discussed the datasets that are accessible for this cause and the application of deep learning models to the situation at hand. The various architectures, tools, algorithms, and datasets employed in several research projects in the area of facial emotion were

examined. Finally, we addressed the current problems and upcoming developments in the field of emotion recognition. The research generally found that transfer learning models are currently leading in the architectures or FER systems since they produce better results. To get better outcomes, several researchers have been seen focusing on enhancing feature extraction and preprocessing methods. Last but not least, researchers saw an exponential increase in multimodal systems when adding additional emotional cues like verbal and electrocardiographic (ECG) cues to create a more precise emotion recognition system. Although facial expression recognition has advanced significantly, the algorithms are still restricted to static photos, which is advantageous in the ideal scenario. However, because human emotions may be so nuanced, methods for capturing subtle, continuous, and ongoing facial expressions are needed in real-world situations. Because there are so many training samples for particular emotions, it is also noted that these emotions are simpler to learn.

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